

LECTURE NOTES MATH254

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1. LECTURE 6

In this class we introduce fraction fields of integral domains.

1.1. Fraction fields of integral domains. Recall that the rational numbers can be constructed from the integers by forming fractions. We want to generalize this construction to all integral domains. To this end, we make the following definition:

Definition 1.1. *Let R be an integral domain. We say that a field F is a fraction field for R if there is an injective map $\psi : R \rightarrow F$ such that if $D = \text{Im}(\psi)$ then any element $x \in F$ can be written as:*

$$(1.1) \quad x = \frac{a}{b},$$

where $a \in D$ and $b \in D \setminus 0$.

Remark 1. We are going to construct a field of fractions for every integral domain R . The construction of a fraction field that we will use is quite standard, and sometimes authors refer to the field obtained by this construction as **the** fraction field. However, it is often useful to have this slightly more general definition of a fraction field.

Remark 2. In some sense the definition of the fraction field F of an integral domain R is completely analogous to how we define $\mathbb{Q}$ given $\mathbb{Z}$. We can consider the map $\psi : \mathbb{Z} \rightarrow \mathbb{Q}$ by $\psi(n) = n/1 = n$, and since any element of $\mathbb{Q}$ is a quotient of two integers we see that $\mathbb{Q}$ is a fraction field for $\mathbb{Z}$.

Lemma 1.1. *Let R be an integral domain and F a fraction field for R . Let L be any field and $\phi : R \rightarrow L$ any (non-zero) homomorphism of rings, then there is a unique homomorphism $\eta : F \rightarrow L$ such that $\eta(\psi(x)) = \phi(x)$ for all $x \in R$.*

Remark 3. This property of a fraction field F is usually called a (or the) **universal property** of the fraction field. As we will see in the next lemma, the universal property is going to guarantee that if an integral domain has a fraction field, then it is unique up to isomorphism (later we will also show that the fraction field exists).

Proof. Let $D = \text{Im}(\psi)$. We define $\eta(a/b) = \phi(\psi^{-1}(a))/\phi(\psi^{-1}(b))$ (note that we can make sense of $\psi^{-1}(a), \psi^{-1}(b)$ since $a, b \in D = \text{Im}(\psi)$ and ψ is injective). First, we must show that η is well-defined. Note that if $a/b = a'/b'$ then $ab' = a'b \in D$ so, since

ψ is injective, we obtain $\phi(\psi^{-1}(ab')) = \phi(\psi^{-1}(a'b))$ and using that $\psi^{-1} : D \rightarrow R$ and ϕ are both ring homomorphisms we have:

$$\begin{aligned}\phi(\psi^{-1}(ab')) &= \phi(\psi^{-1}(a))\phi(\psi^{-1}(b')), \\ \phi(\psi^{-1}(a'b)) &= \phi(\psi^{-1}(a'))\phi(\psi^{-1}(b))\end{aligned}$$

which after dividing (noting that $\phi(\psi^{-1}(b)), \phi(\psi^{-1}(b')) \neq 0$, why is this? Prove it!) implies that:

$$(1.2) \quad \frac{\phi(\psi^{-1}(a))}{\phi(\psi^{-1}(b))} = \frac{\phi(\psi^{-1}(a'))}{\phi(\psi^{-1}(b'))}.$$

This proves that $\eta : F \rightarrow L$ is a well-defined map. Clearly $\eta(\psi(x)) = \phi(x)$. Next we show that η is a field homomorphism. First, note that $\psi(0_R) = 0_F$ since $\psi(0_R) = \psi(0_R + 0_R) = \psi(0_R) + \psi(0_R)$ so after cancelling one $\psi(0_R)$ on both sides we obtain $\psi(0_R) = 0_F$. It follows that $\psi(1_R) \neq 0_F$ since ψ is injective and $0_R \neq 1_R$. Let $x = \psi(1_R)$ then $x^2 = \psi(1_R)\psi(1_R) = \psi(1_R \cdot 1_R) = \psi(1_R) = x$ or after dividing by x , recalling that $x \neq 0$, we have $x = 1_F$. Similarly we see that $\phi(1_R) = 1_L$. Since $\psi(1_R) = 1_F$ we have $\eta(1_F) = \phi(\psi^{-1}(1_F)) = \phi(1_R) = 1_L$. It remains to show that η respects both addition and multiplication. That η respect multiplication is straightforward since ϕ and ψ respect multiplication (check this!). To prove that η respect addition we note that:

$$\begin{aligned}\eta\left(\frac{a}{b} + \frac{a'}{b'}\right) &= \eta\left(\frac{ab' + ba'}{bb'}\right) = \frac{\phi(\psi^{-1}(ab' + ba'))}{\phi(\psi^{-1}(bb'))} = \frac{\phi(\psi^{-1}(ab')) + \phi(\psi^{-1}(ba'))}{\phi(\psi^{-1}(bb'))} = \\ &= \frac{\eta(ab') + \eta(ba')}{\eta(bb')} = \frac{\eta(a)\eta(b') + \eta(b)\eta(a')}{\eta(b)\eta(b')} = \\ &= \frac{\eta(a)}{\eta(b)} + \frac{\eta(a')}{\eta(b')} = \eta\left(\frac{a}{b}\right) + \eta\left(\frac{a'}{b'}\right)\end{aligned}$$

where we have used that ψ and ϕ are ring homomorphisms, and therefore respect addition, and that η respects multiplication. It follows that η is indeed a field homomorphism. Finally, we show that η is unique with the property that $\eta(\psi(x)) = \phi(x)$. However, note that if $\eta'(\psi(x)) = \phi(x)$ then for any $a, b \in D$, $b \neq 0$, we have:

$$(1.3) \quad \eta'\left(\frac{a}{b}\right) = \frac{\eta'(a)}{\eta'(b)} = \frac{\eta'(\psi(\psi^{-1}(a)))}{\eta'(\psi(\psi^{-1}(b)))} = \frac{\phi(\psi^{-1}(a))}{\phi(\psi^{-1}(b))} = \frac{\eta(a)}{\eta(b)} = \eta\left(\frac{a}{b}\right)$$

where the second to last equality use the definition of η . This shows that $\eta' = \eta$ since any element of F is a fraction of elements in D . $\blacksquare$

Lemma 1.2. *Let R be an integral domain. If F and F' are two fraction fields for R , then F and F' are isomorphic as fields. Moreover, if $\psi : R \rightarrow F$ and $\psi' : R \rightarrow F'$ are the injective homomorphisms in the definition of a fraction field, then the isomorphism between F and F' , $\phi : F \rightarrow F'$ is unique if we add the extra assumption that $\phi(\psi(x)) = \psi'(x)$. In particular, if R has a fraction field then it is unique up to isomorphism.*

Proof. Let F, F' be two fraction fields for R with corresponding injective homomorphisms $\psi : R \rightarrow F$ and $\psi' : R \rightarrow F'$. By the previous lemma we obtain two unique

homomorphisms $\phi : F \rightarrow F'$ and $\eta : F' \rightarrow F$ with the properties that $\phi(\psi(x)) = \psi'(x)$ and $\eta(\psi'(x)) = \psi(x)$. It follows that $\phi(\eta(\psi'(x))) = \phi(\psi(x)) = \psi'(x)$ which, since $\phi \circ \eta$ is a field homomorphism (check this!), implies:

$$(1.4) \quad \phi \circ \eta \left(\frac{\psi'(x)}{\psi'(y)} \right) = \frac{\psi'(x)}{\psi'(y)}$$

which, since any element of F' can be written like this, implies that $\phi \circ \eta = \text{id}_{F'}$. Similarly we see that $\eta \circ \phi = \text{id}_F$. It follows that η and ϕ are inverses, so F and F' are isomorphic. Since the η and ϕ were unique with the properties $\phi(\psi(x)) = \psi'(x)$ and $\eta(\psi'(x)) = \psi(x)$, the uniqueness of ϕ (and $\eta = \phi^{-1}$) follows. ■

1.2. Construction of fraction fields. We have seen that if fraction fields exists for some integral domain R , then there is a unique fraction field up to isomorphism. Our next goal is to prove that a fraction field exists for every integral domain. To this end, let R be an integral domain. We define $R^* = R \setminus 0$. Let $\sim$ be the relation on $R \times R^*$ defined by:

$$(1.5) \quad (a, b) \sim (a', b') \text{ if } ab' = a'b.$$

Lemma 1.3. *The relation $\sim$ on $R \times R^*$ is an equivalence relation.*

Proof. It is clear that $(a, b) \sim (a, b)$ since $ab = ab$. If $(a, b) \sim (a', b')$ then $ab' = a'b$ which implies $a'b = ab'$, which in turn means that $(a', b') \sim (a, b)$. It remains to show that if $(a, b) \sim (a', b')$ and $(a', b') \sim (a'', b'')$ then $(a, b) \sim (a'', b'')$. Assume that $ab' = ba'$, $a'b'' = b'a''$, then:

$$(1.6) \quad ab''b' = (ab')b'' = ba'b'' = b(a'b'') = bb'a'' = (ba'')b'$$

or equivalently:

$$(1.7) \quad [ab'' - ba'']b' = 0$$

which, since $b' \neq 0$ and R has no zero divisors, implies that:

$$(1.8) \quad ab'' - ba'' = 0, \quad ab'' = ba''$$

or $(a, b) \sim (a'', b'')$. ■

Definition 1.2. *We define the set $F = (R \times R^*) / \sim$ where $\sim$ is the equivalence relation 1.5. We will write an element in F as either $[a, b]$ or as a/b , $(a, b) \in R \times R^*$.*

Note that we have a map $\psi : R \rightarrow F$ defined by $\psi(x) = [x, 1]$. If $\psi(x) = \psi(y)$ then $(x, 1) \sim (y, 1)$ or $x \cdot 1 = y \cdot 1$, but then $x = y$ so ψ is an injective map. We are going to prove that F is a field, and that ψ is an (as we have already seen) injective homomorphism. This then proves that F is a fraction field as in Definition 1.1.

Lemma 1.4. *We have two well-defined binary operations on F given by:*

$$(1.9) \quad [a, b] + [c, d] = [ad + cb, bd],$$

$$(1.10) \quad [a, b] \cdot [c, d] = [ac, bd].$$

Proof. We need to prove that the two binary operations in the lemma are well-defined. To see the second one, note that if $[a, b] = [a', b']$, and $[c, d] = [c', d']$ then:

$$(1.11) \quad acb'd' = (ab')(cd') = (a'b)(c'd) = a'c'bd$$

which proves that $[ac, bd] = [a'c', b'd']$. That is, the second binary operation is well-defined. To see that the first is well-defined, assume again that $[a, b] = [a', b']$ and $[c, d] = [c', d']$. We have:

$$(ad + cb)b'd' = adb'd' + cbb'd' = (ab')dd' + (cd')bb' = a'bdd' + c'dbb' = (a'd' + c'b')bd$$

so the first binary operation is also well-defined. ■

It remains to prove that F is a field with the binary operations defined in Lemma 1.4, and that any element of F is a fraction of elements in the image of $\psi(x) = [x, 1]$. That is, we must show that $(F, +)$ is an abelian group, that $\cdot$ is associative, that $\cdot$ is commutative, that the distributive law holds, and that every $a \in F \setminus 0$ is a unit. We will not prove all of this, but leave some of the assertions as an exercise.

Lemma 1.5. *The set F with binary operations given in Lemma 1.4 is a field.*

Proof. We begin by proving that both binary operations are commutative. Note that:

$$(1.12) \quad [a, b] \cdot [c, d] = [ac, bd] = [ca, db] = [c, d] \cdot [a, b],$$

and:

$$(1.13) \quad [a, b] + [c, d] = [ad + bc, bd] = [cb + da, db] = [c, d] + [a, b].$$

It remains to prove the following points:

- (i) Addition is associative,
- (ii) the element $[0, 1] \in F$ is an additive identity,
- (iii) for $[a, b]$ we have an additive inverse $[-a, b]$,
- (iv) multiplication is associative,
- (v) the distributive law holds in F ,
- (vi) the element $[1, 1]$ is a multiplicative identity,
- (vii) if $[a, b] \in F$ is such that $[a, b] \neq [0, 1]$ then $[a, b]$ has a multiplicative inverse.

We will only prove the second to last and last assertion and leave the remainder as an exercise (try to do this!). Let $[a, b] \in F$, then:

$$(1.14) \quad [a, b] \cdot [1, 1] = [a \cdot 1, b \cdot 1] = [a, b],$$

so $[1, 1]$ is a multiplicative identity. To prove that last claim, note that if $[a, b] \in F$ and $a \neq 0$ then:

$$(1.15) \quad [b, a] \cdot [a, b] = [ab, ab].$$

We claim that $(ab, ab) \sim (1, 1)$ (note that $b \neq 0$ so $ab \neq 0$ since R has no zero divisors) which proves that $[b, a] \cdot [a, b] = [1, 1]$. To see this simply note that $ab \cdot 1 = ab \cdot 1$ so $(ab, ab) \sim (1, 1)$ or $[ab, ab] = [1, 1]$. This proves that $[a, b]$ has a multiplicative inverse if $a \neq 0$. We claim that $[0, b] = [0, 1]$ for all $b \neq 0$. Equivalently we claim that $(0, b) \sim (0, 1)$ which follows since $0 \cdot 1 = 0 = 0 \cdot b$. It follows that $[a, b]$ has a

multiplicative inverse whenever $[a, b] \neq [0, 1]$ and since $[0, 1]$ is the additive inverse this shows that every non-zero element in F has a multiplicative inverse. ■

Lemma 1.6. *The map $\psi : R \rightarrow F$ defined by $\psi(x) = [x, 1]$ is an injective homomorphism with the property that any $[a, b] \in R$ can be written as a quotient of elements in the image of ψ .*

Proof. We have already seen that ψ is injective. To see that it is a homomorphism, note that:

$$(1.16) \quad \psi(xy) = [xy, 1] = [xy, 1 \cdot 1] = [x, 1] \cdot [y, 1] = \psi(x)\psi(y),$$

$$(1.17) \quad \psi(x + y) = [x + y, 1] = [x \cdot 1 + y \cdot 1, 1 \cdot 1] = [x, 1] + [y, 1] = \psi(x) + \psi(y).$$

That is, ψ respects addition and multiplication so it is a homomorphism. Finally, we need to show that any element in F can be written as a quotient of elements in the image of ψ . Let $[a, b] \in F$ be arbitrary. We note that $[a, 1] \in \text{Im}(\psi)$, $[b, 1] \in \text{Im}(\psi)$ and that the multiplicative inverse of $[b, 1]$ is $[1, b]$. So the product $[a, 1] \cdot [1, b]$ is a quotient of elements in the image of ψ . On the other hand we calculate:

$$(1.18) \quad [a, 1] \cdot [1, b] = [a \cdot 1, 1 \cdot b] = [a, b]$$

and since $[a, b] \in F$ was an arbitrary element of F this finishes the proof of the lemma. ■

By combining Lemmas 1.2, 1.5, and 1.6 we obtain the following theorem on the existence and uniqueness of fraction fields.

Theorem 1.1. *Let R be an integral domain. Then R has a fraction field F , and this fraction field is unique up to isomorphism.*

1.3. An concrete calculation of a fraction field. The construction of a fraction field that we gave in Lemma 1.5 is quite abstract, but it has the advantage that it works for every integral domain R . However, in practice, if we have a little more information about the integral domain R , we can often get more explicit descriptions of fraction fields. In this last section we will give a concrete description of the fraction field of $\mathbb{Z}[i]$ ¹. Let $i \in \mathbb{C}$ be the imaginary unit (i.e., i satisfies $i^2 = -1$). We define:

$$(1.19) \quad \mathbb{Z}[i] = \{a + ib \mid a, b \in \mathbb{Z}\},$$

with binary operations given by normal addition and multiplication. Note that:

$$(1.20) \quad (a + ib) + (c + id) = (a + c) + i(b + d),$$

$$(1.21) \quad (a + ib) \cdot (c + id) = ac + iad + ibc + i^2bd = (ac - bd) + i(ad + bc)$$

and since $a + c, b + d, ac - bd, ad + bc \in \mathbb{Z}$ addition and multiplication are well-defined binary operations on $\mathbb{Z}[i]$. Moreover, the fact that addition and multiplication in $\mathbb{C}$ makes $\mathbb{C}$ a ring is also readily seen to imply that $\mathbb{Z}[i]$ with these operations is a ring. To see that $\mathbb{Z}[i]$ is an integral domain, note that $1 \in \mathbb{Z}[i]$ and that multiplication in $\mathbb{Z}[i]$ is commutative (since multiplication in $\mathbb{C}$ is commutative). It remains to show that $\mathbb{Z}[i]$ does not have zero divisors. Let $0 \neq a + ib \in \mathbb{Z}[i]$ be an arbitrary non-zero

¹The ring $\mathbb{Z}[i]$ is usually called the *Gaussian integers*.

element and let $c + id \in \mathbb{Z}[i]$ be such that $(a + ib)(c + id) = 0$. Our aim is to show that $c + id = 0$ this then implies that $\mathbb{Z}[i]$ has no zero divisors. Note that:

$$(1.22) \quad 0 = (a - ib) \cdot 0 = (a - ib)(a + ib)(c + id) = (a^2 + b^2)(c + id)$$

or after dividing by $a^2 + b^2$ on both sides (note that $a^2 + b^2 \neq 0$ since $a + ib \neq 0$) we obtain:

$$(1.23) \quad 0 = c + id$$

which is what we wanted to show.

Since $\mathbb{Z}[i]$ is an integral domain, it has some fraction field by Theorem 1.1. We will now try to give a concrete description of this fraction field. We begin by defining $\mathbb{Q}[i]$ as:

$$(1.24) \quad \mathbb{Q}[i] = \{x + iy \mid x, y \in \mathbb{Q}\},$$

this is a ring, which can be seen in very much the same way as we see that $\mathbb{Z}[i]$ is a ring. It is also immediate that $\mathbb{Q}[i]$ is a commutative ring with unity since multiplication in $\mathbb{C}$ is commutative, and $1 = 1 + i \cdot 0 \in \mathbb{Q}[i]$. We claim that $\mathbb{Q}[i]$ is in fact a field. To see this, we need to show that any $0 \neq x + iy \in \mathbb{Q}[i]$ has a multiplicative inverse in $\mathbb{Q}[i]$. Fortunately, we know what the multiplicative inverse of $x + iy$ is in $\mathbb{C}$, it is given by $1/(x + iy)$, so we only need to show that the complex number $1/(x + iy)$ is an element of $\mathbb{Q}[i]$. To see this we calculate:

$$\begin{aligned} \frac{1}{x + iy} &= \frac{x - iy}{x - iy} \cdot \frac{1}{x + iy} = \frac{x - iy}{(x - iy)(x + iy)} = \frac{x - iy}{x^2 + y^2} = \\ &\quad \left(\frac{x}{x^2 + y^2} \right) + i \left(\frac{-y}{x^2 + y^2} \right) \end{aligned}$$

but $0 \neq x^2 + y^2$ is a rational number since $x, y \in \mathbb{Q}$ so we obtain:

$$(1.25) \quad \frac{1}{x + iy} = \left(\frac{x}{x^2 + y^2} \right) + i \left(\frac{-y}{x^2 + y^2} \right) \in \mathbb{Q}[i].$$

This proves that $\mathbb{Q}[i]$ have multiplicative inverses for all non-zero elements $x + iy$, so $\mathbb{Q}[i]$ is indeed a field.

We claim that the field $\mathbb{Q}[i]$ is in fact a fraction field for the integral domain $\mathbb{Z}[i]$. We begin by defining a map $\psi : \mathbb{Z}[i] \rightarrow \mathbb{Q}[i]$ by $\psi(a + ib) = a + ib$ (which is a well-defined map since $\mathbb{Z}[i] \subset \mathbb{Q}[i]$). Note that ψ is injective. The map ψ is also a ring homomorphism since addition and multiplication in $\mathbb{Z}[i]$ and $\mathbb{Q}[i]$ are both defined by using addition and multiplication in $\mathbb{C}$. So we have an injective homomorphism $\psi : \mathbb{Z}[i] \rightarrow \mathbb{Q}[i]$ whose image is $\mathbb{Z}[i] \subset \mathbb{Q}[i]$. It remains to show that any element of $\mathbb{Q}[i]$ can be written as a quotient of elements in $\mathbb{Z}[i]$. Let $x + iy \in \mathbb{Q}[i]$ be an arbitrary element and write:

$$(1.26) \quad x = \frac{a}{N}, \quad y = \frac{b}{N}$$

for some integers $a, b, N \in \mathbb{Z}$ where $N \geq 1$ (note that this is always possible, if we have $x = p/q$ and $y = p'/q'$ then we can choose $a = pq'$, $b = p'q$ and $N = qq'$). We

can write:

$$(1.27) \quad x + iy = \frac{a}{N} + i \cdot \frac{b}{N} = \frac{a + ib}{N}$$

but $a + ib \in \mathbb{Z}[i]$, and $N = N + i \cdot 0 \in \mathbb{Z}[i]$, and since $x + iy \in \mathbb{Q}[i]$ was arbitrary we see that any element of $\mathbb{Q}[i]$ can be written as a quotient of elements in $\mathbb{Z}[i]$. We have proved the following:

Theorem 1.2. *The field $\mathbb{Q}[i]$ is a fraction field for the integral domain $\mathbb{Z}[i]$.*

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